

Mark schemes

Q1.

$$3x \frac{dy}{dx} + 3y - 4y \frac{dy}{dx} = 0$$

attempt implicit differentiation

M1

product

A1

chain

A1

constant

B1

$$\frac{dy}{dx} = -\frac{3}{2}$$

CSO

A1

ALTERNATIVE METHOD

$$x = \frac{2}{3}y + \frac{4}{3y}$$

*solve for x = expression in y and
differentiate with respect to y*

(M1)

$$\frac{dx}{dy} = \frac{2}{3} - \frac{4}{3y^2}$$

(A1A1)

$$y = 1, \frac{dx}{dy} = \frac{2}{3} - \frac{4}{3}$$

substitute y = 1

(M1)

$$\frac{dx}{dy} = -\frac{3}{2}$$

CSO

(A1)

[5]

Q2.

(a) $V = x(9 - 3x)^2$

*Attempt at V in terms of x (condone slip
when rearranging formula for $y = 9 - 3x$)
or $(9 - 3x)^2 = 81 - 54x + 9x^2$*

M1

$$V = x(81 - 54x + 9x^2)$$

$$= 81x - 54x^2 + 9x^3$$

AG; no errors in algebra

A1

2

(b) (i) $\frac{dV}{dx} = 81 - 108x + 27x^2$

One term correct

M1

Another correct

A1

All correct (no + c etc)

A1

$$= 27(x^2 - 4x + 3)$$

CSO; all algebra and differentiation correct

A1

4

(ii) $(x - 3)(x - 1)$ or $(27x - 81)(x - 1)$ etc

"Correct" factors or correct use of formula

M1

$$\Rightarrow x = 1, 3$$

*SC: B1, B1 For $x = 1$, $x = 3$ found by
inspection (provided no other values)*

A1

2

(c) $\frac{d^2V}{dx^2} = -108 + 54x$ (condone one slip)

$\frac{dV}{dx}$
ft their (may have cancelled 27 etc)

M1

CSO; all differentiation correct

A1

2

(d) (i) $x = 3 \Rightarrow \frac{d^2V}{dx^2} = 54; \quad x = 1 \Rightarrow \frac{d^2V}{dx^2} = -54$

$\frac{d^2V}{dx^2}$
ft their and their **two** x -values

B1ft

1

(ii) $(x =) 1$ (gives maximum value)

$\frac{d^2V}{dx^2} < 0$
Provided their

E1

1

(iii) $V_{\max} = 36$
 CAO

B1

1

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[13]

Q3.

(a) $\sqrt{x^3} = x^{\frac{3}{2}}$

OE; accept 'k = 1.5'

B1

1

(b) (i) $\frac{dy}{dx} = 2x - \frac{3}{2}x^{\frac{1}{2}}$

At least one index reduced by 1 and no
 term of the form $\sqrt{ax^2}$.

M1

For $2x$

B1

For $-1.5x^{0.5}$. Ft on ans (a) non-integer k

A1F

3

(ii) When $x = 4, y = 8$

B1

$$y'(4) = ;$$

Attempt to find $\frac{dy}{dx}$ when $x = 4$

M1

$$= 2(4) - 1.5(\sqrt{4}) = 5$$

Ft on one earlier error provided noninteger powers in (a) and (b) (i)

A1F

$$\text{Tangent: } y - 8 = 5(x - 4)$$

$$y - y(4) = y'(4)[x - 4] \text{ OE}$$

m1

$$y = 5x - 12$$

CSO; must be $y = 5x - 12$

A1

5

[9]

Q4.

(a) (i) $h = 0.5$
 Integral = $h/2 \{ \dots \}$
PI

B1

$$\{ \dots \} = f(0) + 2\left[f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{3}{2}\right)\right] + f(2)$$

OE summing of areas of the four traps.

M1

$$\{ \} = 1 + 2[\sqrt{6} + 6 + 6\sqrt{6}] + 36$$

$$= 1 + 2[2.449.. + 6 + 14.6969..] + 36$$

Condone 1 numerical slip. Accept 3sf values if not exact.

A1

$$= 37 + 2 \times 23.146.. = 83.292...$$

$$\text{Integral} = 0.25 \times 83.292.. = 20.8 \text{ (3sf)}$$

CAO; must be 20.8

A1

4

- (ii) Relevant trapezia drawn on a copy of given graph
Accept single trapezium with its sloping side above the curve

M1

{Approximation is an} overestimate
Dep. on 4 trapezia with each of their upper vertices lying on the curve

A1

2

- (b) (i) Stretch (I) in x -direction (II)
Need (I) and one of (II), (III)
M0 if more than one transformation

M1

(scale factor) $\frac{1}{3}$ (III)

A1

2

- (ii) $6^{3x} = 84$
PI

M1

$$\log_{10} 6^{3x} = \log_{10} 84$$

Take logs of both sides of $a^x = b$, PI by 'correct' value(s) later or $3x = \log_6 84$

M1

$$3x \log_{10} 6 = \log_{10} 84$$

Use of $3 \log 6^{3x} = 3x \log 6$ OE

or $3x = \log_6 84$ seen

m1

$$x = \frac{\lg 84}{3 \lg 6}$$

$$x = 0.82429\dots = 0.824 \text{ (to 3dp)}$$

*Must see that logs have been used before
any of the last 3 marks are awarded in
(b)(ii). Condone > 3dp*

A1

4

(c) $f(x) = 6^{x-1} - 2$

B1 For either $6^{x-1} + 2$ or for $6^{x+1} - 2$

B2,1

2

[14]

Q5.

(a) $\frac{dy}{dx} = -x^2 \sin 2x + \cos 2x$

product rule $kx \sin 2x \pm \cos 2x$

M1

no further incorrect working

A1

2

(b) (i) $-2\alpha \sin 2\alpha + \cos 2\alpha = 0$

*replacing $x = \alpha$ and writing equation
equal to zero (at any line)*

M1

$$\left. \begin{array}{l} 2\alpha \sin 2\alpha = \cos 2\alpha \\ 2\alpha \tan 2\alpha = 1 \end{array} \right\} \text{either}$$

$$2\alpha \tan 2\alpha - 1 = 0$$

AG; CSO

A1

2

(ii) $\left. \begin{array}{l} f(0.4) = 0.2 \\ f(0.5) = -0.6 \end{array} \right\} \text{awrt} \quad \text{o.e}$

(0.9's unsubstantiated scores M0)

M1

Change of sign $\therefore 0.4 < \alpha < 0.5$

A1

2

(iii) $2x \tan 2x = 1$

$$\left. \begin{array}{l} \tan 2x = \frac{1}{2x} \\ 2x = \tan^{-1}\left(\frac{1}{2x}\right) \end{array} \right\} \text{either}$$

$$x = \frac{1}{2} \tan^{-1}\left(\frac{1}{2x}\right)$$

AG; CSO

B1

1

(iv) $x_1 = 0.4$
 $x_2 = 0.4480\dots$
 $x_2 = 25.7$

$x_3 = 0.4200\dots$
 $= 0.42$

M1

A1

2

(c) $y = x \cos 2x$

$$\left. \begin{array}{l} u = x \quad du = 1 \\ dv = \cos 2x \quad v = \frac{\sin 2x}{2} \end{array} \right\}$$

$\left. \begin{array}{l} \text{differentiate one term} \\ \text{integrate one term} \end{array} \right\} \text{ must be } k \sin 2x$

M1

*correct substitution of their values into
 parts formula using $u = x$*

m1

$$\int = \frac{x \sin 2x}{2} - \int \frac{\sin 2x}{4} (dx)$$

$$= \left[\frac{x \sin 2x}{2} + \frac{\cos 2x}{4} \right]_{(0)}^{(0.5)}$$

A1

$$= \left(\frac{\sin 1}{4} + \frac{\cos 1}{4} \right) - \left(\frac{\cos 0}{4} \right)$$

*correctly substituting values from
previous 2 method marks*

m1

$$= 0.0954$$

AWRT

A1

5

[14]

Q6.

(a) $\frac{dx}{dt} = 4$ $\frac{dy}{dt} = -\frac{1}{2t^2}$

differentiate. 4; at⁻² seen

M1

both derivatives correct

A1

$$\frac{dy}{dx} = -\frac{1}{2t^2} \times \frac{1}{4}$$

use chain rule candidates' $\frac{dy}{dt} / \frac{dx}{dt}$

M1

$$t = \frac{1}{2} \quad \frac{dy}{dx} = -\frac{1}{2}$$

CSO

A1

ALTERNATIVE METHOD:

$$y = \frac{2}{x-3} - 1 \quad \text{and differentiate}$$

differentiate expression in y and x

M1

$$\frac{dy}{dx} = \frac{-2}{(x-3)^2}$$

correct

A1

$$x = 5$$

$$\frac{dy}{dx} = \frac{-2}{(5-3)^2}$$

find and therefore use x (and y)

m1

$$\frac{dy}{dx} = -\frac{1}{2}$$

A1

4

(b) gradient of normal = 2

F if gradient $\neq \pm 1$

B1F

$$(x, y) = (5, 0) \quad \frac{y}{x-5} = 2$$

calculate and use (x, y) on normal

M1

F on gradient of normal ACF

A1F

3

$$(c) \quad x - 3 = 4t \quad \text{or} \quad y + 1 = \frac{1}{2t}$$

$$\text{or} \quad t = \frac{x-3}{4} \quad \text{or} \quad \frac{1}{t} = 2(y+1)$$

B1

$$(x-3)(y+1) = 2$$

eliminate t ; allow one error

M1

$$y = \frac{1}{2(x-3)} - 1 \quad \text{ACF}$$

accept

SC allow marks for part (c) if done in part (a)

A1

3

[10]

Q7.

(a) (i) $2x^2 + 2xh + 4xh \quad (= 54)$

Attempt at surface area (one slip)

M1

$$\Rightarrow x^2 + 3xh = 27$$

AG CSO

A1

2

(ii) $h = \frac{27 - x^2}{3x}$ or $h = \frac{9}{x} - \frac{x}{3}$ etc
Any correct form

B1

1

(iii) $V = 2x^2h = 18x - \frac{2x^3}{3}$
AG (watch fudging) condone omission of brackets

B1

1

(b) (i) $\frac{dV}{dx} = 18 - 2x^2$

One term correct "their" V

M1

All correct unsimplified $18 - 6x^2/3$

A1

2

(ii) Sub $x = 3$ into their $\frac{dV}{dx}$

Or attempt to solve their $\frac{dV}{dx} = 0$

M1

Shown to equal 0 plus **statement** that this implies a stationary point if verifying

CSO Condone $x = \pm 3$ or $x = 3$ if solving

A1

2

(c) $\frac{d^2V}{dx^2} = -4x$

(= -12)

FT their $\frac{dV}{dx}$

B1 ✓

$\frac{d^2V}{dx^2} < 0$

at stationary point $\Rightarrow$ maximum

FT their second derivative conclusion

If "their" $\frac{d^2y}{dx^2} > 0 \Rightarrow$ minimum etc

E1 ✓

2

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[10]

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Q8.

(a) (i) $B(0, 5)$

B1

Area AOB $\frac{1}{2} \times 1 \times 5$

Condone slip in number or a minus sign

M1

$= 2\frac{1}{2}$

A1

3

(ii) $\frac{3x^6}{6} + \frac{2x^2}{2} + 5x$ or $\frac{x^6}{2} + x^2 + 5x$ (may have + c or not)

Raise one power by 1

M1

One term correct

A1

All correct unsimplified

A1

3

(iii) Area under curve = $\int_{-1}^0 f(x) dx$
Correctly written or $F(0) - F(-1)$ correct

B1

$[0] - \left[\frac{1}{2} + 1 - 5 \right]$

*Attempt to sub limit(s) of -1 (and 0)
 Must have integrated*

M1

Area under curve $3\frac{1}{2}$
CSO (no fudging)

A1

Area of shaded region = $3\frac{1}{2} - 2\frac{1}{2} = 1$

FT their integral and triangle (very generous)

B1 ✓

4

(b) (i) $\frac{dy}{dx} = 15x^4 + 2$

One term correct

M1

All correct (no +c etc)

A1

when $x = -1$, gradient = 17

CSO

A1

3

(ii) $y = \text{"their gradient"} (x + 1)$

*Must be finding **tangent** – not normal
any form e.g. $y = 17x + 17$*

B1 ✓

1

[14]

Q9.

(a) (i) $y = x + 1 + 4x^{-2} \Rightarrow \frac{dy}{dx} = 1 - 8x^{-3}$

M1

Power $p \rightarrow p - 1$

(A1 if $1 + ax^n$ with $a = -8$ or $n = -3$)

A2,1,0

3

(ii) $1 - 8x^{-3} = 0$

Puts c's $\frac{dy}{dx} = 0$

M1

$x^3 = 8$

Using $x^{-k} = \frac{1}{x^k}$ to reach $x^a = b$, $a > 0$ or correct use of logs.

m1

$x = 2$

A1

When $x = 2$, $y = 4$

A1ft

4

(iii) At (1, 6), $\frac{dy}{dx} = 1 - 8 = -7$
Attempt to find $y'(1)$

M1

$$\text{Gradient of normal} = \frac{1}{7}$$

Use of or stating $m \times m' = -1$

M1

$$\text{Equation of normal is } y - 6 = m[x - 1]$$

m numerical

M1

$$y - 6 = \frac{1}{7}(x - 1)$$

$$\left\{ \frac{y-6}{x-1} = \frac{1}{7}; 7y = x + 41 \right\}$$

OE ft on c's answer for (a)(i) provided at least A1 given in (a)(i) and previous 3M marks awarded

Alft

4

(b) (i)

$$\int x \left(1 + \frac{4}{x^2} \right) dx =$$

$$\dots\dots\dots = \frac{x^2}{2} + x - 4x^{-1} \{+c\}$$

One of three terms correct.

M1

One of three terms correct.

*For A2 need all three terms as printed or better
(A1 if 2 of 3 terms correct)*

A2,1,0

3

(ii)

$$\{\text{Area} = \} \int_1^4 x + 1 + \frac{4}{x^2} dx =$$

$$\left[\frac{x^2}{2} + x - \frac{4}{x} \right]_1^4 = (8 + 4 - 1) - \left(\frac{1}{2} + 1 - 4 \right)$$

*Dealing correctly with limits; $F(4) - F(1)$
(must have integrated)*

M1

$$= 13.5$$

A1

2

[16]

Q10.

(a) (i) $y = (4x^2 + 3x + 2)^{10}$

$$\frac{dy}{dx} = 10(4x^2 + 3x + 2)^9 (8x + 3)$$

M1

For $f(x)^9$ where $f(x) \neq k$ and is linear

A1

2

(ii) $y = x^2 \tan x$

Product rule

M1

$$\frac{dy}{dx} = x^2 \sec^2 x + 2x \tan x$$

A1

2

(b) (i) $x = 2y^3 + \ln y$

$$\frac{dx}{dy} = 6y^2 + \frac{1}{y}$$

B1

1

(ii) At (2,1)

$$\frac{dx}{dy} = 6 + 1 = 7$$

M1

$$\frac{dy}{dx} = \frac{1}{7}$$

May be implied

A1 ✓

$$(y-1) = \frac{1}{7}(x-2)$$

OE

A1

3

[8]

Q11.

(a) (i) $\int (4 - e^{2x}) dx$

$$= 4x - \frac{1}{2} e^{2x} (+c)$$

4x

B1

$$- \frac{1}{2} e^{2x}$$

B1

2

(ii) $\int_0^{\ln 2} (4 - e^{2x}) dx = \left[4x - \frac{1}{2} e^{2x} \right]_0^{\ln 2}$

$$= \left[4 \ln 2 - \frac{1}{2} e^{2 \ln 2} \right] - \left[(0) - \frac{1}{2} (e^0) \right]$$

Substitute both ln 2 and 0 correctly into an integrated expression

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M1

$$= 4 \ln 2 - 2 + \frac{1}{2}$$

Convincing

$$= 4 \ln 2 - \frac{3}{2}$$

AG

A1

2

(b) (i) $x = 0$

$$y = 4 - 1 = \underline{\underline{3}}$$

B1

1

(ii) At B , $y = 0$

$$4 - e^{2x} = 0$$

Or reverse argument

M1

$$e^{2x} = 4$$

$$x = \ln 2$$

AG

A1

2

(c) $\frac{dy}{dx} = -2e^{2x}$

B1

$$x = \ln 2, \text{ Gradient} = -2e^{2\ln 2} = -8$$

$$x = \ln 2 \text{ into } ke^{2x}$$

M1

$$\text{Gradient normal} = \frac{1}{-8} = -\frac{1}{8e^{2\ln 2}}$$

OE

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A1

Equation $y = -\frac{1}{8}x - \frac{1}{8}\ln 2$

OE

A1

4

(d) When $x = 0$

$$y = -\frac{1}{8}\ln 2$$

*Attempt to integrate their line and substitute
 $x = 0, \ln 2$*

M1

Area $\Delta = \frac{1}{16} (\ln 2)^2$ condone – ve sign = 0.03
 $\frac{1}{2} (their y) \times \ln 2$

A1 ✓

Total area $4 \ln 2 - \frac{3}{2} + \frac{1}{16} (\ln 2)^2 = 1.30$ AWRT
 CSO

A1

3

[14]

Q12.

(a) (i) $\frac{dx}{dt} = 2, \frac{dy}{dt} = -8t$
 CAO

B1, B1

2

(ii) $\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{-8t}{2} = -4t$

M1

Chain rule in correct form
 ft on sign coefficient errors (**not** power of t)

A1F

2

(b) $m_T = -4, m_N = \frac{1}{4}$
 $\frac{dy}{dx}$
 ft on $\frac{dy}{dx}$ if f(t)

B1F,
 B1F

$x = 3 \quad y = -3$

$\frac{y-3}{x-3} = \frac{1}{4} \Rightarrow \frac{y+3}{x-3} = \frac{1}{4}$

Use candidate's (x, y) and m_N

M1

Any correct form; ISW; CAO

A1

4

(c) $t = \frac{x-1}{2}$

M1

$$y = 1 - 4 \left(\frac{x-1}{2} \right)^2$$

Substitute for t

Simplification not required but CAO

Or equivalent methods / forms:

$$y = 2x - x^2, \quad t^2 = \frac{1-y}{4},$$

$$\left(\frac{x-1}{2} \right)^2 = \frac{1-y}{4}$$

M1A1

3

[11]

Q13.

(a) (i) $t^3 - 52t + 96$

one term correct

M1

another term correct

A1

all correct (no + c etc)

A1

3

(ii) $3t^2 - 52$

ft one term correct

M1

ft all "correct"

A1 ✓

(b) $\frac{dy}{dt} = 8 - 10t + 96$

substitute $t = 2$ into their $\frac{dy}{dt}$

M1

$= 0 \Rightarrow$ stationary value
CSO; shown = 0 + statement

A1

Substitute $t = 2$ into $\frac{d^2y}{dt^2}$ ($= -40$)
any appropriate test, e.g. $y'(1)$ and $y'(3)$

M1

$\frac{d^2y}{dt^2} < 0 \Rightarrow$
 max value
all values (if stated) must be correct

A1

(c) Substitute $t = 1$ into their $\frac{dy}{dt}$
must be their $\frac{dy}{dt}$ NOT $\frac{d^2y}{dt^2}$

M1

Rate of change = $45 \text{ (cm s}^{-1}\text{)}$
ft their $y'(1)$

A1 ✓

(d) Substitute $t = 3$ into their $\frac{dy}{dt}$
interpreting their value of $\frac{dy}{dt}$

M1

$(27 - 156 + 96 = 33 < 0)$

$\Rightarrow$ decreasing when $t = 3$

allow increasing if their $\frac{dy}{dt} > 0$

E1 ✓

2

[13]

Q14.

(a) $y_P = 4$

B1

1

(b) $y = 1 + \frac{2}{x} + \frac{2}{x} + \frac{4}{x^2}$

(B1 if only one error in the expansion)
For B2 the last line of the candidate's solution must be correct

B2,1,0

$y = 1 + 4x^{-1} + 4x^{-2}$

2

(c) $\frac{dy}{dx} = -4x^2 - 8x^{-3}$

Index reduced by 1 after differentiating x
to a negative power

M1

At least 1 term in x correct ft on expn

A1ft

CSO Full correct solution. ACF

A1

3

(d) When $x = 2$, $\frac{dy}{dx} = -4 \times 2^{-2} - 8 \times 2^{-3}$

Attempt to find $y'(2)$.

M1

Gradient = $-1 - 1 = -2$

AG (be convinced-no errors seen)

A1

(e) $-2 \times m' = -1$

$m_1 \times m_2 = -1$ OE stated or used. PI

M1

$y - 4 = m(x - 2)$

C's y_p from part (a) if not recovered;
 m must be numerical.

M1

$y - 4 = \frac{1}{2}(x - 2)$

Ft on candidate's y_p from part (a) if not recovered.

A1ft

$x - 2y + 6 = 0$

CAO Must be this or $0 = x - 2y + 6$

A1

4

[12]

Q15.

(a) $y = \ln x$

penalise + c once on 1 (a) or 2 (a)

$\frac{dy}{dx} = \frac{1}{x}$

B1

1

(b) $y = (x + 1) \ln x$

$\frac{dy}{dx} = (x + 1) \times \frac{1}{x} + \ln x$

product rule

M1

A1

2

(c) $y = (x + 1) \ln x$

$$\frac{dy}{dx} = \frac{1}{x} + 1 + \ln x$$

$$x = 1: \quad \frac{dy}{dx} = 1 + 1 = 2$$

substitute $x = 1$ into their $\frac{dy}{dx}$

M1

$$\text{Grad normal} = -\frac{1}{2}$$

use of $m_1 m_2 = -1$

M1

CSO

A1

$$y = -\frac{1}{2}(x-1)$$

OE

A1

4

[7]

Q16.

(a) (i) $y = (x^2 - 3)e^x$

$$\frac{dy}{dx} = (x^2 - 3)e^x + 2xe^x$$

product rule

M1

A1

2

(ii) $\frac{d^2 y}{dx^2} = (x^2 - 3)e^x + 2xe^x + 2xe^x + 2e^x$

product rule from their $\frac{dy}{dx}$

M1

A1

2

(b) (i) $\frac{dy}{dx} = 0$

$$\Rightarrow e^x(x^2 + 2x - 3) = 0$$

$$e^x f(x) = 0 \text{ from } \frac{dy}{dx} = 0$$

M1

$$e^x(x + 3)(x - 1) = 0$$

attempt at factorising or use of formula

m1

$$\therefore x = -3, 1$$

first correct solution

A1

second correct solution, and no others

SC No working shown:

$$x = -3 \quad B2, \quad x = 1 \quad B2$$

A1

4

(ii) $x = -3 \quad y'' = -4e^x \text{ max } (-0.2)$

Condone slip

M1

$$x = 1 \quad y'' = 4e^x \text{ min } (10.9)$$

A1

2

[10]

Q17.

(a) $x = 1, 5a^2 - a - 4 = 0$

condone y for a

M1

$$(5a + 4)(a - 1) = 0, a = 1$$

AG – be convinced, both factors seen

$$\text{or } a = -\frac{4}{5} \text{ or } 1 \Rightarrow a = 1$$

A0 for 2 positive roots

(substitute (1, 1) $\Rightarrow$ 5 = 5 no marks)

A1

2

(b) $\frac{dy}{dx} + 4$

$\frac{dy}{dx}$
(Ignore ' $\frac{dy}{dx}$ ' if not used, otherwise loses final A1)

B1B1

$$= 10xy^2 + 10x^2y \frac{dy}{dx}$$

attempt product rule, see two terms added

M1

$\frac{dy}{dx}$
chain rule, $\frac{dy}{dx}$ attached to one term only

M1

condone 5×2 for 10

A1

$$x=1, y=1 \quad \frac{dy}{dx} + 4 = 10 + 10 \frac{dy}{dx}$$

$\frac{dy}{dx}$
two terms, or more, in $\frac{dy}{dx}$

M1

$$\frac{dy}{dx} = -\frac{6}{9} = \left(-\frac{2}{3}\right)$$

CSO

A1

7

Alt (for last two marks)

$$\frac{dy}{dx} = \frac{10xy^2 - 4}{1 - 10x^2y}$$

$\frac{dy}{dx}$
find $\frac{dy}{dx}$ in terms of x, y and substitute $x = 1$,
 $y = 1$ must be from expression with two terms



or more in $\frac{dy}{dx}$

(M1)

$$(1, 1) \Rightarrow \frac{10-4}{1-10} = -\frac{6}{9}$$

(A1)

(c) $\frac{y-1}{x-1} = -\frac{2}{3}$ (OE)

ft on gradient
ISW after any correct form

B1F

1

[10]

Q18.

(a) (i) $\frac{dV}{dt} = 2t^5 - 8t^3 + 6t$
One term correct unsimplified

M1

Further term correct unsimplified

A1

All correct unsimplified (no + c etc)

A1

3

(ii) $\frac{d^2V}{dt^2} = 10t^4 - 24t^2 + 6$
One term FT correct unsimplified

M1

CSO. *All correct simplified*

A1

2

(b) Substitute $t = 2$ into their $\frac{dV}{dt}$

M1

$$= 64 - 64 + 12 = 12$$

CSO. Rate of change of volume is $12 \text{ m}^3 \text{ s}^{-1}$

A1

2

(c) (i) $t = 1 \Rightarrow \frac{dV}{dt} = 2 - 8 + 6$

Or putting their $\frac{dV}{dt} = 0$

M1

$= 0 \Rightarrow$ Stationary value

CSO. Shown to $= 0$ **AND** statement
(If solving equation must obtain $t = 1$)

A1

2

(ii) $t = 1 \Rightarrow \frac{d^2V}{dt^2} = -8$

Sub $t = 1$ into their second derivative or
equivalent full test.

M1

Maximum value

Ft if their test implies minimum

A1 ✓

2

EXAM PAPERS PRACTICE

[11]

Q19.

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(a) $y_D = 3 + 1 = 4$ or $y_C = 12 - 8 = 4$

Attempt at either y coordinate

M1

Area $ABCD = 3 \times 4 = 12$

A1

2

(b) (i) $x^3 - \frac{x^4}{4} \quad (+ C)$

Increase one power by 1

M1

One term correct unsimplified

A1

All correct unsimplified (condone no +C)

A1

3

- (ii) Sub limits -1 and 2 into their (b)(i) ans

May use both $-1, 0$ and $0, 2$ instead

M1

$$[8 - 4] - \left[-1 - \frac{1}{4} \right] = 5 \frac{1}{4}$$

A1

Shaded area = "their" (rectangle – integral)

Alt method: difference of two integrals

M1

$$= 12 - 5 \frac{1}{4} = 6 \frac{3}{4}$$

CSO. Attempted M2, A2

A1

4

(c) (i) $\frac{dy}{dx} = 6x - 3x^2$

One term correct

M1

All correct (no +C etc)

A1

2

- (ii) When $x = 1, y = 2$ when $x = 1,$

May be implied by correct tgt equation

B1

$$\frac{dy}{dx} = 3 \text{ as 'their' grad of tgt}$$

Ft their derivative when $x = 1$

M1ft

Tangent is $y - 2 = 3(x - 1)$

Any correct form $y = 3x - 1$ etc

A1

3

- (iii) Decreasing when $\frac{dy}{dx} = 6x - 3x^2 < 0$
Watch no fudging here!!
May work backwards in proof.

M1

$$3(2x - x^2) < 0 \Rightarrow x^2 - 2x > 0$$

AG (be convinced no step incorrect)

A1

2

- (d) Two critical points 0 and 2
Marked on diagram or in solution

M1

$x > 2$, $x < 0$ ONLY
 or M1 A0 for $0 < x < 2$ or $0 > x > 2$
SC B1 for $x > 2$ (or $x < 0$)

A1

2

[18]

Q20.

$$y'(x) = 16 - x^{-2}$$

One term correct

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Both correct

A1

$$y'(x) = 16 - \frac{1}{x^2}$$

$$x^{-2} = \frac{1}{x^2} \quad \text{OE PI}$$

B1

$$y'(x) = 0 \Rightarrow 16x^2 = 1;$$

c's $y'(x) = 0$ and one relevant further step

M1

$$\Rightarrow x = \pm \frac{1}{4}$$

Both answers required.

A1

5

[5]

Q21.

(a) $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} - 3$

One term correct

M1

Both correct

A1

2

(b) (i) When $x = 0$, $\frac{dy}{dx} = -3$
Ft provided answer < 0.

Eqn of tangent at O is $y = -3x$
OE Ft on $y'(0)$

B1F ✓

B1F ✓

2

(ii) At $(9,0)$ $\frac{dy}{dx} = \frac{3}{2}(9)^{\frac{1}{2}} - 3$

Attempt to find $y'(9)$

M1

Eqn tangent at A is $y - 0 = y'(9)[x - 9]$
OE

m1

$$\Rightarrow y = \frac{3}{2}(x - 9) \Rightarrow 2y = 3x - 27$$

CSO. AG

A1

3

(iii) Eliminating $y \Rightarrow -6x = 3x - 27$

OE method to one variable

(eg $2y = -y - 27$)

M1

$$9x = 27 \Rightarrow x = 3$$

*[A1F for each coordinate; only ft on $y = kx$ tangent
in (b)(i) for $k < 0$]*

A1F

When $x = 3, y = -9\}. \quad P(3, -9)\}$

A1F

3

(c) $\int \left(x^{\frac{3}{2}} - 3x \right) dx = \frac{2}{5} x^{\frac{5}{2}} - \frac{3x^2}{2} (+c)$

One power correct

M1

Condone absence of "+ c" and unsimplified forms

A2,1,0

3

(d) $\int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx =$

PI

B1

$$= \frac{2}{5} \times 9^{\frac{5}{2}} - \frac{3}{2} \times 9^2 - 0$$

Correct use of limits following integration

M1

$$= -24.3$$

$$\text{Area of triangle } OPA = \frac{1}{2} \times 9 \times |y_P|$$

M1

$$\text{Sh.Area} = \frac{1}{2} \times 9 \times |y_P| - \left| \int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx \right|$$

OE

M1

$$= 40.5 - 24.3 = 16.2$$

A1

5

[18]

Q22.

(a) $\frac{dy}{dt} = \frac{-2}{t^2} \quad \frac{dx}{dt} = -4$

M1A1

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{\frac{dx}{dt}} = \frac{1}{2t^2}$$

Use chain rule

m1

Follow on use of chain rule (if f(t))

A1F

Or eliminate t : M1 y = f(x) attempt to differentiate M1A1 chain rule

A1F reintroduce t

4

(b) $t = 2 \quad m_T = \frac{1}{8}$

follow on gradient (possibly used later)

B1F

$$x = -5 \quad y = 2$$

B1

$$y - 2 = \frac{1}{8} (x + 5)$$

Their (x, y), m

M1

$$x - 8y + 21 = 0$$

Ft on (x, y) and m

A1F

4

(c) $x - 3 = -4t$ $y - 1 = \frac{2}{t}$
PI

M1

$(x - 3)(y - 1) = -4t \times \frac{2}{t} = (-8)$
Attempt to eliminate t

M1

AG convincingly obtained

A1

3

[11]

Q23.

(a) $\frac{dy}{dx} = -10x^4$
kx⁴ condone extra term

M1

Correct derivative unsimplified

A1

2

(b) When $x = 1$, gradient = -10
FT their gradient when $x = 1$

B1 ✓

Tangent is

Attempt at y & tangent (not normal)

M1

$y - 5 = -10(x - 1)$ or $y + 10x = 15$ etc
CSO Any correct form

A1

3

(c) When $x = -2$ $\frac{dy}{dx} = -160$ (or < 0)

Value of their $\frac{dy}{dx}$ when $x = -2$

B1 ✓

($\frac{dy}{dx} < 0$ hence) y is **decreasing**

ft Increasing if their $\frac{dy}{dx} > 0$

E1 ✓

2

[7]

Q24.

(a) (i) $\frac{dy}{dx} = 3x^2 - 20x + 28$

One term correct

M1

Another term correct

A1

All correct (no + c etc)

A1

3

(ii) Their $\frac{dy}{dx} = 0$ for stationary point
Or realising condition for stationary point

M1

$$(x - 2)(3x - 14) = 0$$

Attempt to solve using formula/ factorise

m1

$$\Rightarrow x = 2$$

A1

or $x = \frac{14}{3}$

Award M1, A1 for verification that $x = 2 \Rightarrow \frac{dy}{dx} = 0$

then may earn m1 later

A1

4

(b) (i) $\frac{x^4}{4} - \frac{10x^3}{3} + 14x^2 + c$

One term correct unsimplified

M1

Another term correct unsimplified

A1

All correct unsimplified (condone missing + c)

A1

3

(ii) $\left[\frac{81}{4} - 90 + 126 \right] \quad (-0)$

Attempt to sub limit 3 into their (b)(i)

M1

$= 56\frac{1}{4}$

AG Integration, limit subtraction all correct

A1

2

(iii) Area of triangle = $31\frac{1}{2}$

Correct unsimplified $\frac{1}{2} \times 21 \times 3$

B1

Shaded Area = $56\frac{1}{4} - \text{triangle area}$

M1

$= 24\frac{3}{4}$

Or equivalent such as $\frac{99}{4}$

B1

3

[15]

Q25.

(a) (i) When $x = 4$, $\frac{dy}{dx} = 3(2) + \frac{16}{16} - 7 = 0$

AG Be convinced

B1

1

(ii) $\frac{16}{x^2} = 16x^{-2}$

Accept $k = -2$

B1

1

(iii) $\frac{d^2y}{dx^2} = 3 \times \frac{1}{2} x^{-\frac{1}{2}} + 16 \times (-2)x^{-3} - 0$

A power decreased by 1

M1

$\frac{d^2y}{dx^2} = \frac{3}{2} x^{-\frac{1}{2}} ; -32x^{-3}$

candidate's negative integer k [-1 for >2 term(s)]

A1;
A1 ✓

3

(iv) When $x = 4$, $\frac{d^2y}{dx^2} = \frac{3}{4} - \frac{32}{64} = \frac{1}{4}$

Attempt to find $y''(4)$ reaching as far as two simplified terms

M1

Minimum since $y''(4) > 0$

candidate's sign of $y''(4)$

E1 ✓

2

[Alternative: Finds the sign of $y'(x)$ either side of the point where $x = 4$, need evidence rather than just a statement: (M1)

Correct ft conclusion with valid reason E1 ✓]

[In both, condone absent statement $y'(4) = 0$]

(b) (i) At $P(1,8)$, $\frac{dy}{dx} = 3(1)^{\frac{1}{2}} + \frac{16}{1^2} - 7 = 12$
AG Be convinced

B1

1

(ii) Gradient of normal = $-\frac{1}{12}$
Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 8 = m[x - 1]$
Can be awarded even if $m = 12$

M1

$y - 8 = -\frac{1}{12}(x - 1) \Rightarrow 12y - 96 = -x + 1 \Rightarrow 12y + x = 97$
Any correct form of the equation

A1

3

$\int 3x^{\frac{1}{2}} + \frac{16}{x^2} - 7dx = \dots = 3 \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + 16 \frac{x^{-1}}{-1} - 7x + c$

(c) (i) *One power correct.*

M1

*A1 if 2 of 3 terms correct candidate's negative integer k
 Condone absence of "+ c"*

A2,1,0 ✓

3

(ii) $y = 2x^{\frac{3}{2}} - 16x^{-1} - 7x + c$ (*)
 $y =$ candidate's answer to (c)(i) with tidied coefficients and with '+c'. ('y =' PI by next line)

B1 ✓

When $x = 1, y = 8 \Rightarrow 8 = 2 - 16 - 7 + c$
Substitute. (1,8) in attempt to find constant of integration

M1

$$y = 2x^{\frac{3}{2}} - 16x^{-1} - 7x + 29$$

Accept $c = 29$ after (*), including $y =$, stated

A1

3

[17]

Q26.

(a) $x = 1 \quad y^2 - y + 3 - 5 = 0$

M1

$$(y - 2)(y + 1) = 0$$

Attempt to solve quadratic equation with
 $x = 1$

M1

$$y = 2 \quad y = -1$$

A1

3

(b) (i) $2y \frac{dy}{dx} - x \frac{dy}{dx} - y + 6x = 0$
 $+6x; -5 \rightarrow 0$

B1B1

Chain rule

B1

Product rule (M1 two terms)

M1A1

$$6x - y + (2y - x) \frac{dy}{dx} = 0$$

Factorise and obtain answer given

A1

6

Alternative

$$\frac{dy}{dx} (y - x)^2 = (y - x)(0 - 6x)$$

$$5 \rightarrow 0$$

(B1)

$$-6x$$

(B1)

$$- (5 - 3x^2) \left(\frac{dy}{dx} - 1 \right)$$

Recognisable attempt at quotient rule

(M1)

Completely correct OE

(A1)

$$\frac{dy}{dx} \left[(y+x)^2 + (5-3x^2) \right] = (y-x)(-6x) + (5-3x^2)$$

Factorise out $\frac{dy}{dx}$

(A1)

Given answer

Correct answer from correct working

Be convinced

(A1)

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(ii) $(1, 2) \quad \frac{dy}{dx} = -\frac{4}{3}$

Substitute $x = 1$ and one y value from (a)

M1

$$(1, -1) \quad \frac{dy}{dx} = \frac{7}{3}$$

Both; follow on candidates y s

$$\text{OE } -\frac{7}{3}; \text{ 3SF}$$

A1F

2

(iii) $y - 6x = 0$

B1

$$(6x)^2 - x \times 6x + 3x^2 - 5 = 0$$

M1

$$36x^2 - 6x^2 + 3x^2 - 5 = 0$$

$$33x^2 - 5 = 0$$

AG convincingly obtained

A1

3

[14]

Q27.

(a) $\frac{dy}{dx} = 5x^4 - 18x^2 - 3$

Decrease one power by 1

M1

One term correct

A1

All correct

A1

3

(b) (i) Sub $x = 2$ into their $\frac{dy}{dx}$
 $80 - 72 - 3$

M1

Shown to equal 5

AG (be convinced)

A1

2

(ii) Gradient of normal $= -\frac{1}{5} \left(y + \frac{1}{5}x + \dots \right)$
Or $m_1 m_2 = -1$ used or stated

B1

$$y - 3 = -\frac{1}{5}(x - 2)$$

*Trying normal NOT tangent
 or $y = mx + c$ and attempt to find c*

M1

$x + 5y = 17$ (integer coefficients)
Or integer multiple of coefficients

A1

3

(c) Sub $x = 1$ into their $\frac{dy}{dx}$ ($= -16 < 0$)

$$(5 - 18 - 3 = -16) \text{ (Watch } \frac{d^2y}{dx^2} = -16 \text{ !)}$$

M1

Negative value $\Rightarrow$ DECREASING

Correct interpretation of sign of $\frac{dy}{dx}$

E1ft

2

[10]

Q28.

(a) Sides $24 - 2x$, $9 - 2x$
Either correct

B1

$V = x(24 - 2x)(9 - 2x)$
3 sides involving x multiplied together

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M1

$$= 4x^3 - 66x^2 + 216x$$

AG (be convinced)

A1

3

(b) (i) $\frac{dV}{dx} = 12x^2 - 132x + 216$
Power decreased by 1

M1

One term correct

A1

All correct (no +C etc)

A1

3

- (ii) Putting their $\frac{dV}{dx} = 0$ (must see this first)
 Or their $12x^2 - 132x + 216 = 0$
 Or $12(x^2 - 11x + 18) = 0$ or statement

M1

$$\Rightarrow x^2 - 11x + 18 = 0$$

AG (be convinced)

A1

2

- (iii) $(x - 2)(x - 9) = 0$

M1

Factors, comp sq or formulae used (1 slip)
 $x = 2, x = 9$

A1

2

- (iv) Reject $x = 9$, since $9 - 2x < 0$
 $x = 2$ is only possible value

E1

1

- (c) (i) $\frac{d^2V}{dx^2} = 24x - 132$

Differentiating their $\frac{dV}{dx}$ (eg $2x - 11$)

M1

Correct

A1

2

- (ii) $x = 2$ only $\Rightarrow \frac{d^2V}{dx^2} = -84$ (or < 0)

Correct $\frac{d^2V}{dx^2}$ value OE full test.

B1

Maximum value

ft if their test implies minimum

E1 ✓

2

[15]

Q29.

(a) (i) $y = x + 2x^{-1}$

PI by sight of $-2x^{-2}$

B1

$$\frac{dy}{dx} = 1 - 2x^{-2}$$

One term correct

M1

OE

A1

3

(ii) When $x = 2$, $\frac{dy}{dx} = 1 - \frac{2}{4} = \frac{1}{2}$
CSO AG (be convinced)

A1

1

(b) When $x = 2$, $y = 3$
For $y = 3$

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gradient of normal = -2

$$m \times m' = -1 \text{ used}$$

M1

Equation normal $y - 3 = -2(x - 2)$

$$y - 3 = m(x - 2) \text{ OE}$$

M1

Award at 1st correct form

A1

4

[8]

Q30.

(a) (i) $(2 + x)^3 =$

M1

$$(2^3) + 3(2^2)(x) + 3(2)(x^2) + (x^3)$$

A1

$$\dots = 8 + 12x + 6x^2 + x^3 (*)$$

Any valid method; must contain all components

Accept $a = 12$

Accept $b = 6$

A1

3

(ii) $(2 - x)^3 = 8 - 12x + 6x^2 - x^3 (**)$

Clear $x \rightarrow -x$ in (i) OE

M1

ft numerical a and b

A1ft

2

(b) $(2 + x)^3 - (2 - x)^3 = (*) - (**)$

Subtracts the 2 expressions in (a)

M1

$$\dots\dots = 24x + 2x^3$$

CSO AG (be convinced)

A1

2

(c) $\frac{dy}{dx} = 24 + 6x^2$

A power of x decreased by 1

M1

For st. pt. $24 + 6x^2 = 0$

A1

Not possible since $24 + 6x^2 > 0$

Any valid explanation



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